

Papermache: Quantitative Evaluation of Scientific Information Value per Manuscript

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Abstract

We define the mathematical goal and tournament comparison metric for PAPERMACHE, a system evaluating the invariant information value of competing scientific manuscripts on shared theoretical foundations.

1 Goal

Given two scientific manuscripts W_A and W_B addressing the same physical, chemical, or informational domain $\mathcal{D}$, the goal of PAPERMACHE is to compute the asymmetric information value operator:

$$\Delta\mathcal{V}(A, B) = \mathcal{V}(W_A) - \mathcal{V}(W_B) \quad (1)$$

determining which work delivers greater non-tautological constraint on the physical state space per unit of communication overhead.

2 Information Value Metric

2.1 Irreducible Invariant Payload

Let a manuscript W be decomposed into an ordered set of formal propositions $\mathcal{T} = \{T_1, T_2, \dots, T_K\}$. The invariant payload $\Omega(W)$ is:

$$\Omega(W) = \sum_{k=1}^K w(T_k) \cdot \mathbb{I}(T_k \notin \mathcal{K}_0) \quad (2)$$

where $\mathcal{K}_0$ is the prior literature knowledge base, $\mathbb{I}(\cdot)$ is the indicator function, and $w(T_k)$ weights operator assertions:

$$w(T_k) = \begin{cases} 3.0 & \text{Formal Theorem / Conservation Law} \\ 2.5 & \text{Hamiltonian / Variational Algorithm} \\ 2.0 & \text{Asymptotic Scaling Bound } O(f(N)) \\ 1.5 & \text{Quantitative Benchmark Invariant} \\ 0.0 & \text{Tautology / Qualitative Preamble} \end{cases} \quad (3)$$

2.2 Downstream Generative Reach

The generative reach $\Gamma(T_k)$ quantifies the logarithm of the Hilbert space dimension or parameter volume constrained

by invariant T_k :

$$\Gamma(T_k) = \log_2 \left(\frac{\text{Vol}(\mathcal{S}_{\text{unconstrained}})}{\text{Vol}(\mathcal{S}_{\text{constrained}}(T_k))} \right) \quad (4)$$

2.3 Directed Mutual Information via Cross-Perplexity

Under an autoregressive compression policy π_θ , the empirical entropy of manuscript B is:

$$\mathcal{L}(B) = -\frac{1}{|B|} \sum_{t=1}^{|B|} \log_2 \pi_\theta(b_t \mid b_{<t}) \quad (5)$$

The conditional loss of B given prior conditioning on manuscript A is:

$$\mathcal{L}(B \mid A) = -\frac{1}{|B|} \sum_{t=1}^{|B|} \log_2 \pi_\theta(b_t \mid A, b_{<t}) \quad (6)$$

The directed empirical mutual information $\hat{I}(A \rightarrow B)$ is:

$$\hat{I}(A \rightarrow B) = \max(0, \mathcal{L}(B) - \mathcal{L}(B \mid A)) \quad (7)$$

2.4 Boilerplate Fluff Penalty

Let $N_{\text{pulp}}(W)$ denote the token count of discursive rhetoric, boilerplate citations, and non-operative transitional phrases, with total token length $|W|$. The fluff ratio is:

$$\mathcal{P}(W) = \frac{N_{\text{pulp}}(W)}{|W|} \quad (8)$$

2.5 Total Composite Information Value

The global information value $\mathcal{V}(W)$ is:

$$\mathcal{V}(W) = \alpha \cdot \Omega(W) + \beta \cdot \sum_{k=1}^K \Gamma(T_k) - \gamma \cdot \mathcal{P}(W) \quad (9)$$

with normalized scaling parameters $\alpha = 2.0$, $\beta = 0.8$, and $\gamma = 1.5$.

2.6 Minimax Tournament Decision

The net tournament superiority margin is:

$$\Delta\mathcal{V}(A, B) = \mathcal{V}(W_A) - \mathcal{V}(W_B) + \mu \left(\frac{\hat{I}(A \rightarrow B)}{|W_A|} - \frac{\hat{I}(B \rightarrow A)}{|W_B|} \right) \quad (10)$$

where $\mu > 0$ enforces the penalty for asymmetric redundancy.

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